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Large Language Models

Introduction





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Introduction LLMs



Why do LLMs matter?

Introduction

LLMs changed the interface from code to conversation.

- Ask questions in natural language
- Generate code, summaries, emails, reports
- Build AI applications without training a model from scratch

You

Write a Taylor Swift song about generative AI

ChatGPT

(Verse 1)

In a digital world, spinning round and round,
Found a spark of art in the code, a silent sound,
Dreaming in RGB, a canvas so wide,
Generative AI, with you I confide.

I know how to chat!

(Verse 2)

Pixels dance, algorithms in a trance,
Crafting beauty with a binary glance,
From chaos to clarity, from dark to light,
You're drawing a new dawn, taking flight.

(Chorus)

Oh, generative muse, paint me a world,
Where the lines are blurred, and the stories unfurled,
With each iteration, you're breaking the chains,
Taylor-made dreams in digital frames.

(Verse 2)

You weave a symphony from a dataset's soul,
Composing the future, playing an unknown role,
A touch of humanity in the machine's eye,
Generative AI, teach us to fly.



UI/UX

Productivity

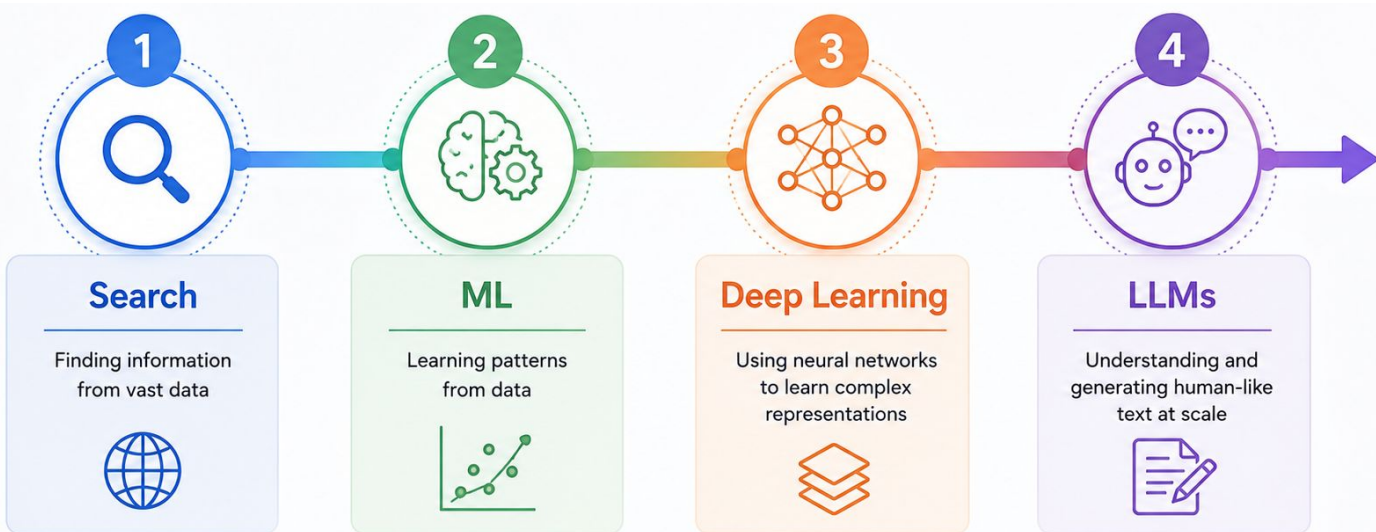
Natural Language



Why Are We Learning LLMs?

Introduction

The Evolution of AI



From finding information → to understanding → to intelligent generation





Why Are We Learning LLMs?

Introduction

**ARTIFICIAL
INTELLIGENCE**



**MACHINE
LEARNING**



**DEEP
LEARNING**



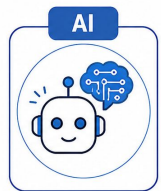
**LARGE
LANGUAGE MODELS**





Where does an LLM fit?

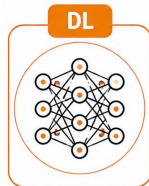
Introduction



Making machines act intelligently



Learning patterns from data



Using neural networks



Deep learning model specialized for language



Systems that generate new content

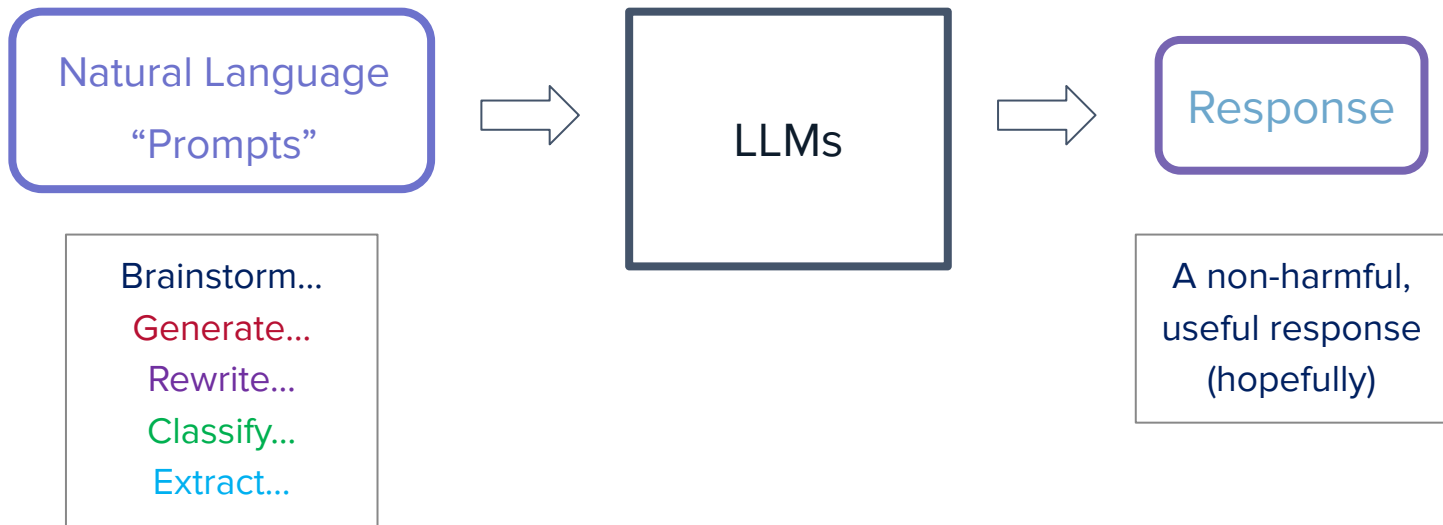
LLMs are a subset of Deep Learning and a major engine behind Generative AI.



What is an LLM?

Introduction

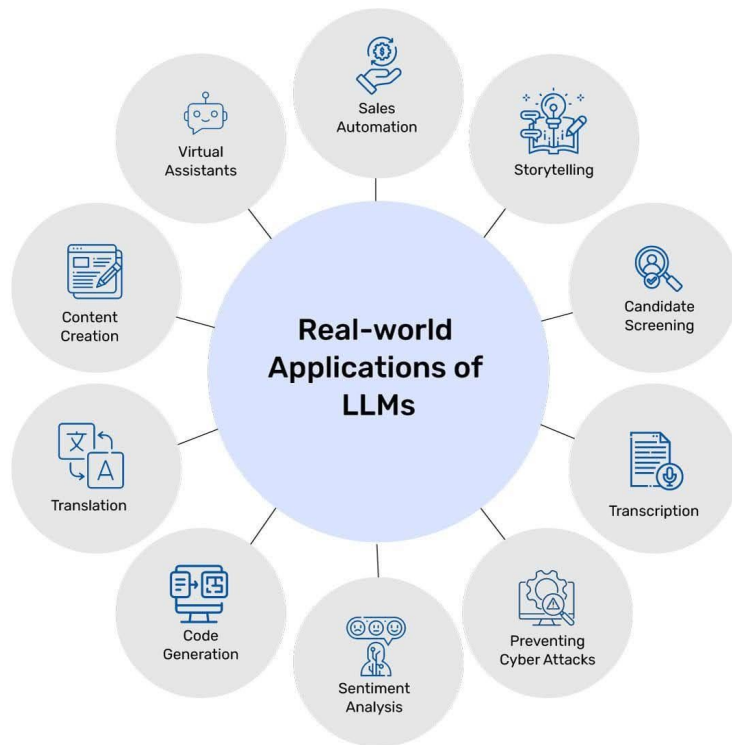
A Large Language Model is a neural network trained on massive amounts of text to predict the next token (word or word-piece) in a sequence.



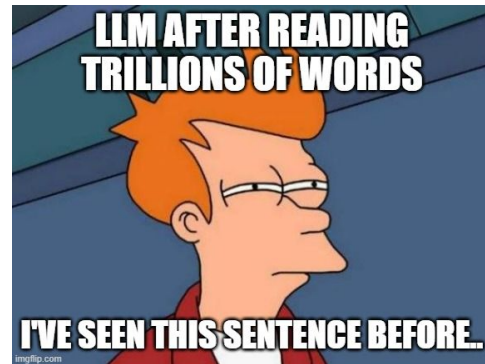


What can an LLM do?

Introduction



<https://www.mindbrowser.com/llm-application-development/>





Why are LLMs different from traditional NLP?

Introduction



NLP

- One model per task
- Requires feature engineering
- Rigid pipelines
- Limited generalization



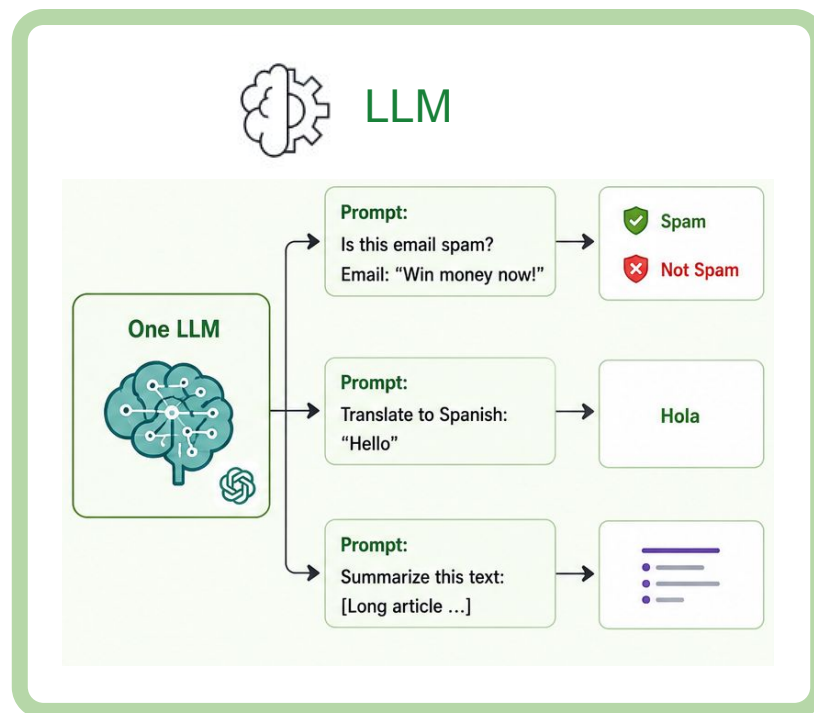
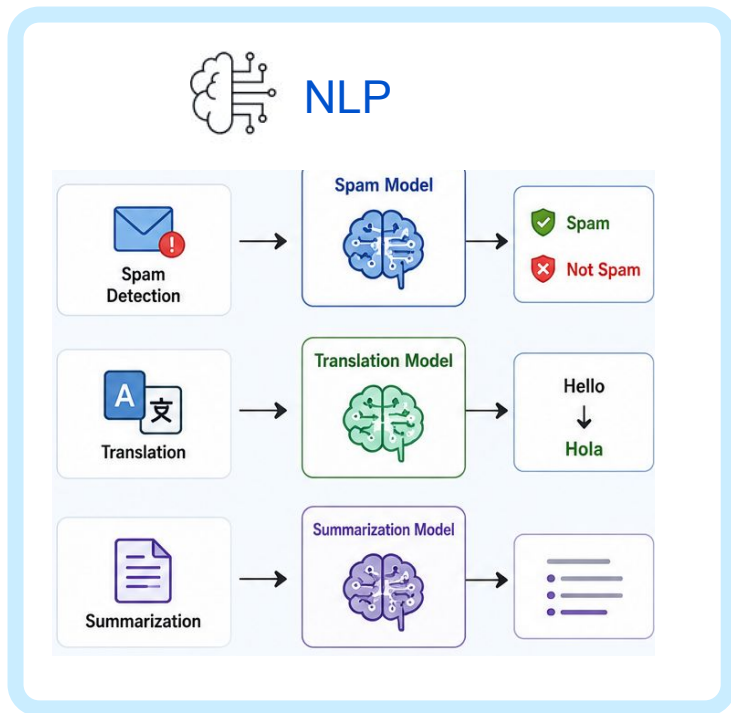
LLM

- One model, many tasks
- Learns representations automatically
- Natural language interface
- Broad general capabilities



Why are LLMs different from traditional NLP?

Introduction





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How LLMs Learn?

LLMs



Self-supervised Learning

Definition

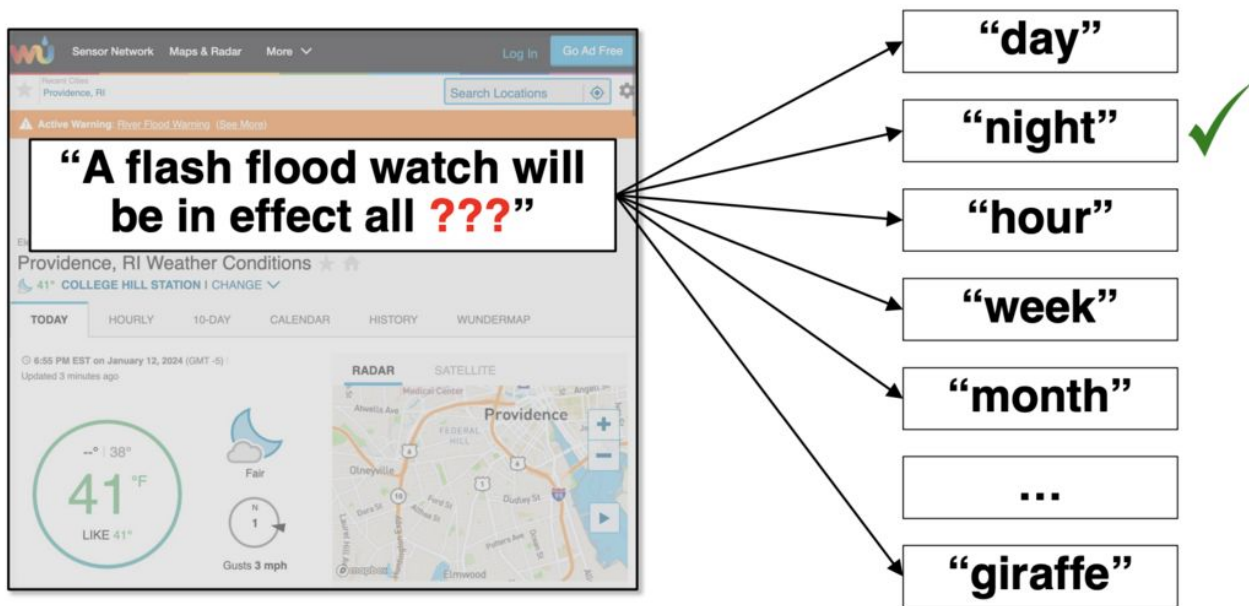
Self-supervised learning (SSL) is a machine learning method where:

- a model trains itself on unlabeled data by creating its own supervised tasks.
- It hides part of the data and tasks the model with predicting the missing pieces.



Self-supervised Learning

How do LLMs learn without Labels?



<https://snorkel.ai/blog/large-language-model-training-three-phases-shape-llm-training/>



Why is this powerful?

Self-supervised Learning

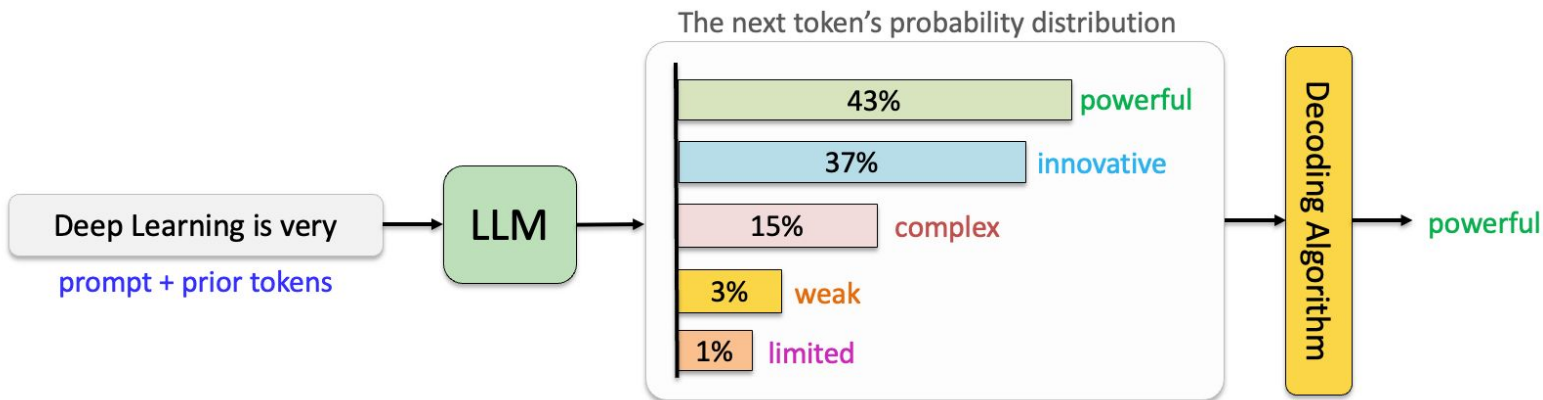
- No manual labeling for billions of sentences
- Can use books, websites, articles, and code
- Scales to internet-sized datasets





Next Token Prediction

Self-supervised Learning



<https://medium.com/@lmpo/mastering-llms-a-guide-to-decoding-algorithms-c90a48fd167b>



Why does it work?

Self-supervised Learning

After seeing trillions of examples, the model learns:

- Grammar
- Facts
- Writing styles
- Reasoning patterns
- Code patterns



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Foundation Models

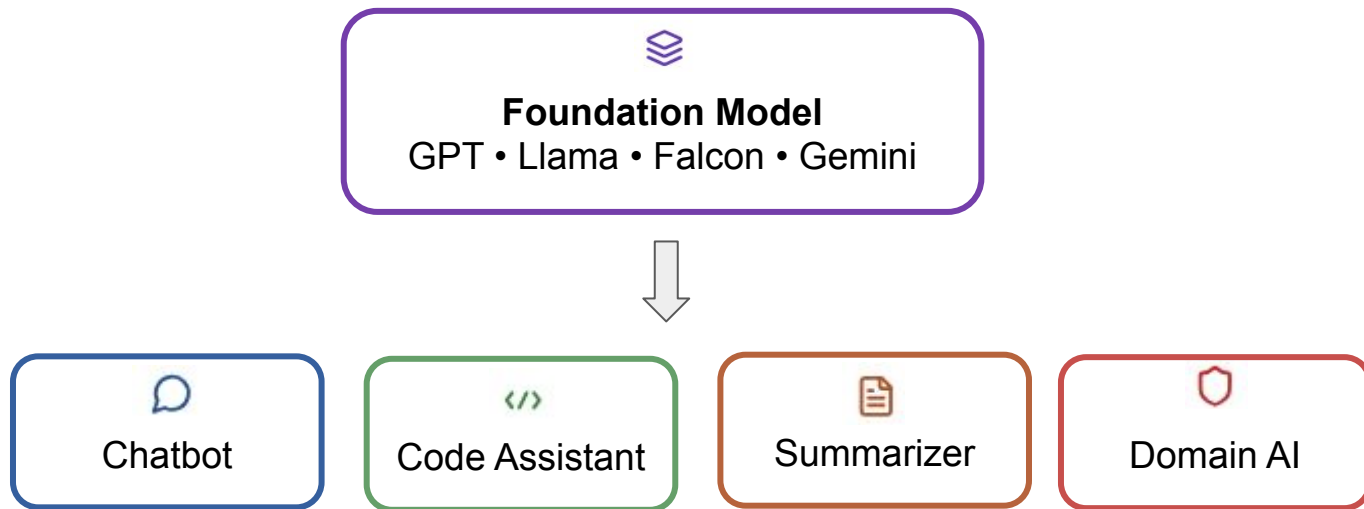
LLMs



Foundation Models

LLM

A large pretrained model that can be adapted to many downstream tasks.



Most companies do not train foundation models. They build applications on top of them.



Open Source Vs Closed Source

Foundation Models

Closed Source	Open Source
GPT-5, Claude	Llama, Falcon, Mistral
API access	Download weights
Managed by provider	Run on your infrastructure
Easier to start	More control and privacy



When should you choose each?

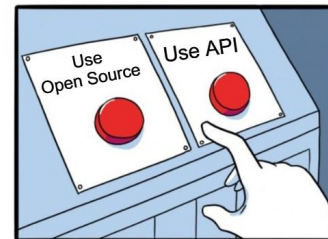
Foundation Models

Use APIs when:

- You want fast development
- You have a small team
- Infrastructure is not your focus

Use Open-source when:

- Data privacy is critical
- Regulations require local deployment
- You need customization



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YAPE-CLARK-TUMBLR



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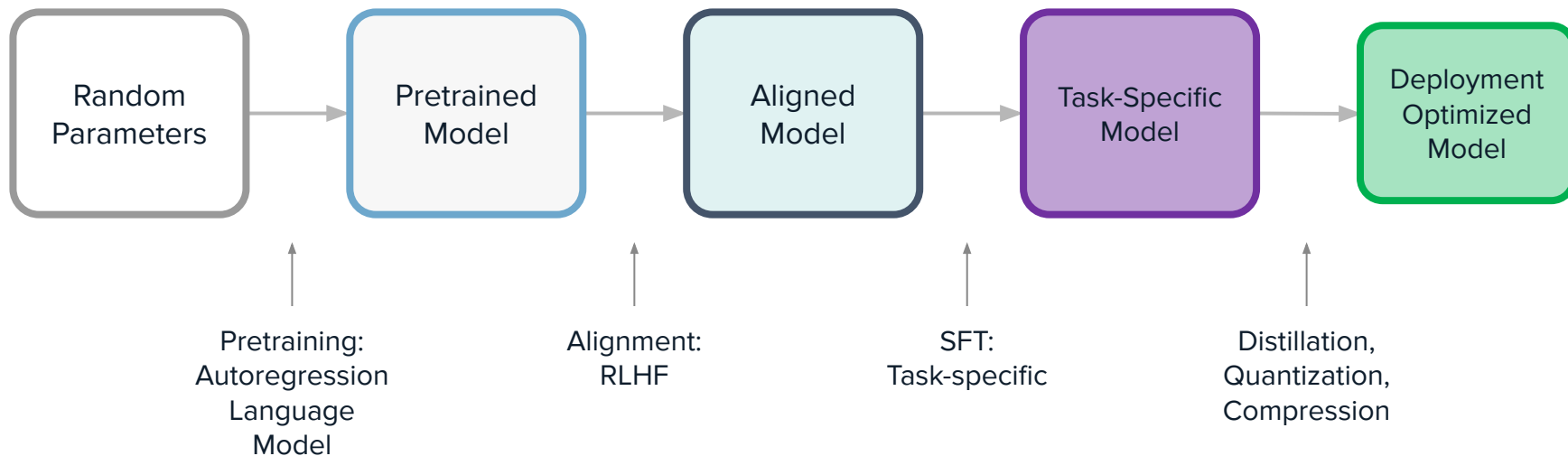
Adapting and Using LLMs

LLMs



LLM Life Cycle

LLMs





How do we actually use an LLM?

LLMs

Good prompts consisting of 3 parts:

Part	Example
Instruction	“Summarize the article”
Context	Paste the article
Output Format	“Return 5 bullet points”

“Summarize the following article for a business executive in 5 bullet points.”



Why are LLMs Useful?

LLMs

Create

- Emails
- Reports
- Presentations
- Marketing copy

Transform

- Summarize
- Translate
- Rewrite
- Extract information

Build

- Code assistants
- Chatbots
- RAG systems
- AI agents

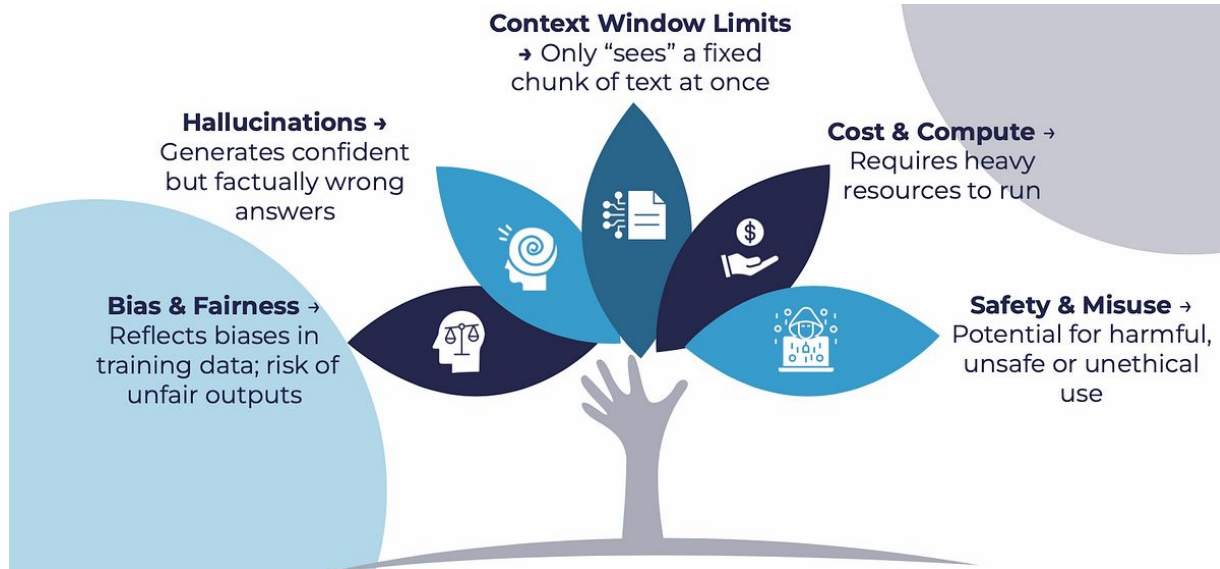
Operate

- Customer support
- Knowledge search
- Compliance assistance
- Internal copilots



Limitations

LLMs



<https://medium.com/@abhishekjunnarkar/limitations-and-risks-of-llms-what-you-as-a-practitioner-should-know-d066a52bcf1a>



Summary

- Explained what Large Language Models (LLMs) are and how they fit within Artificial Intelligence and Generative AI.
- Learned how LLMs are trained using self-supervised learning and next-token prediction.
- Introduced foundation models and compared open-source and closed-source LLMs.
- Explored the LLM lifecycle.
- Learned how prompts allow users to interact with LLMs and perform a wide variety of tasks.
- Examined real-world applications of LLMs across industries.
- Discussed the limitations of LLMs, including hallucinations, bias, privacy concerns, and computational costs.



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Thank You

We'd love to hear from you. If you have any questions,
feel free to reach out to our team anytime!